

05 Design Practice

1 Introduction

This example is very similar to what students can expect on the midterm. They should try to solve the second exercise in 45 minutes, that's the amount of time they will have on the midterm.

On the midterm, students can choose whether to just call algorithmic patterns like `SEARCH(quiver, cond(a) -> a.type = type)`, or implement them themselves. The former is preferred. If they call them, they have to use the correct notation. If they choose to implement them themselves, they have to use enumerators (foreach loops), and have a correct implementation for each. Half-baked implementations or ones that have errors are worth 0 points.

The first assignment is also similar to the midterm, if someone wants to practice they can solve additional ones besides the one assigned to them.

2 Library

A Library has a collection of books (list of Book objects). Each book has an Author (String), pageCount (int), and a genre (String). The genre could be, for example, fiction, non-fiction, mystery, or science-fiction.

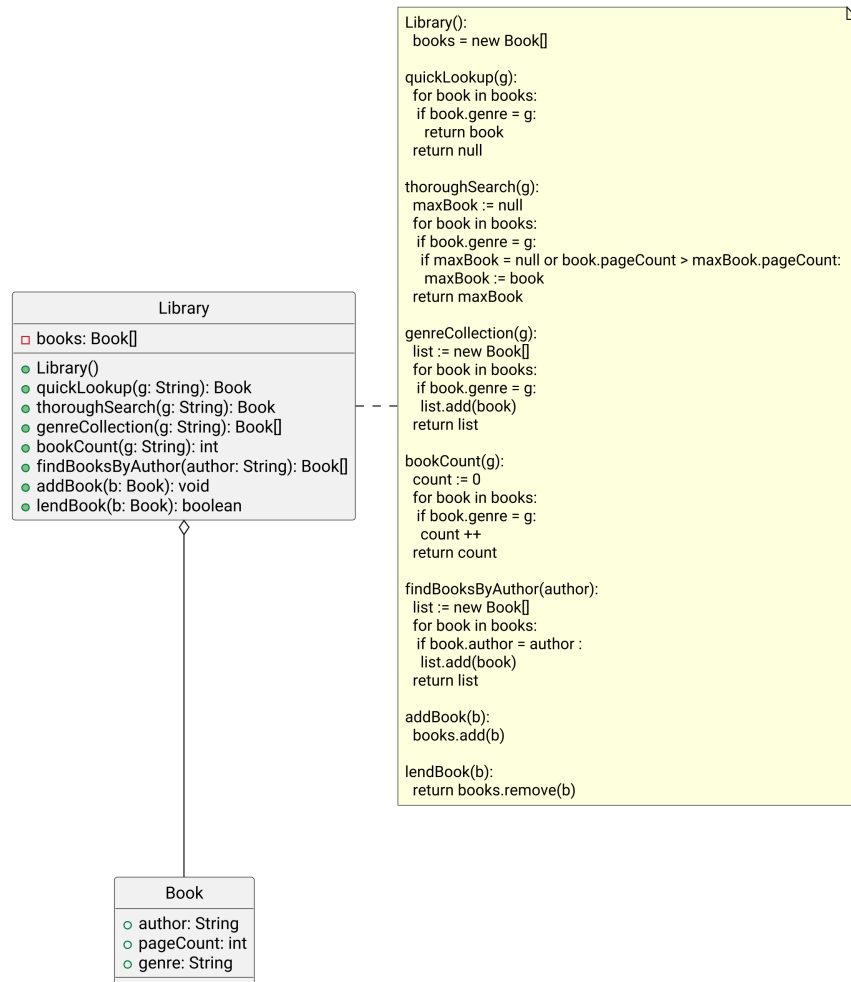
The librarian has different tasks that require them to find books:

- **Quick Lookup:** They retrieve the first book of a given genre.
- **Thorough Search:** They look for a book with the maximum page count in a given genre.
- **Genre Collection:** They return all the books in a given genre.
- **Book Count:** They want to know the number of books of a given genre.
- **FindBooksByAuthor:** They look for all books written by a given author.

New books can be added using the **addBook** method, and books can be lent by the **lendBook** method. We don't model returning books to the library, we just add them again.



Solution without algorithmic patterns:



Solution with algorithmic patterns:

